

MATLAB Simulation Exercise

Statistical Signal Processing II

EKF (EXTENDED KALMAN FILTER)

Group 09

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February 11, 2026

Theoretical Background

In many radar and navigation applications, the objective is to estimate the motion of a moving target using noisy sensor measurements. Typically, the target state consists of its position and velocity components and evolves over time according to a dynamic motion model. When the measurements are nonlinear functions of the state (such as range and bearing), the standard Kalman filter cannot be applied directly. In such cases, the Extended Kalman Filter (EKF) provides an approximate recursive minimum mean-square error (MMSE) estimator by linearizing the nonlinear measurement function.

In this experiment, the target state vector (signal vector) is defined as the position and velocity components as

$$s[n] = \begin{bmatrix} r_x[n] \\ r_y[n] \\ v_x[n] \\ v_y[n] \end{bmatrix}$$

here, $r_x[n]$ and $r_y[n]$ represent the Cartesian coordinates of the target, and $v_x[n]$ and $v_y[n]$ represent the corresponding velocity components.

The target is assumed to follow a constant velocity model, described by the linear state equation:

$$v_x[n] = v_x[n-1] + u_x[n] \quad (1)$$

$$v_y[n] = v_y[n-1] + u_y[n] \quad (2)$$

And at the position at time n is:

$$r_x[n] = r_x[n-1] + v_x[n-1]\Delta \quad (3)$$

$$r_y[n] = r_y[n-1] + v_y[n-1]\Delta \quad (4)$$

These equations can be written compactly in state-space form as

$$\mathbf{s}[n] = A \mathbf{s}[n-1] + \mathbf{u}[n] \quad (5)$$

With the transition matrix

$$A = \begin{bmatrix} 1 & 0 & \Delta & 0 \\ 0 & 1 & 0 & \Delta \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

and process noise vector

$$u[n] = \begin{bmatrix} 0 \\ 0 \\ u_x[n] \\ u_y[n] \end{bmatrix}$$

The corresponding process noise covariance matrix is therefore

$$Q = Eu[n]u^T[n] = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_u^2 & 0 \\ 0 & 0 & 0 & \sigma_u^2 \end{bmatrix}$$

Measurement Model: The sensor provides noisy observations of the range and bearing of the vehicle with respect to the origin. Thus, the observation equation is:

$$x[n] = \begin{bmatrix} \sqrt{r_x^2[n] + r_y^2[n]} \\ \arctan\left(\frac{r_y[n]}{r_x[n]}\right) \end{bmatrix} + \begin{bmatrix} w_r[n] \\ w_\beta[n] \end{bmatrix}$$

$$x[n] = h(s[n]) + w[n]$$

where $w_r[n]$ and $w_\beta[n]$ are independent, zero-mean, white Gaussian measurement noises:

$$w_r[n] \sim N(0, \sigma_r^2)$$

$$w_\beta[n] \sim N(0, \sigma_\beta^2)$$

Hence the measurement noise covariance is

$$C = Ew[n]w^T[n] = \begin{bmatrix} \sigma_r^2 & \mathbf{0} \\ \mathbf{0} & \sigma_\beta^2 \end{bmatrix}$$

Because the measurement function $h(\cdot)$ is nonlinear, the standard Kalman filter cannot be applied directly. EKF applies the Kalman filter recursion by linearizing the nonlinear measurement function $h(\mathbf{s})$ about the current predicted state.

Prediction:

Given the estimate $\hat{\mathbf{s}}[n-1 | n-1]$ and covariance $M[n-1 | n-1]$, the predicted state and covariance are

$$\hat{\mathbf{s}}[n | n-1] = A\hat{\mathbf{s}}[n-1 | n-1] \quad (7)$$

$$M[n | n - 1] = A M[n - 1 | n - 1] A^T + Q \quad (8)$$

Linearization:

The Jacobian matrix of the measurement function evaluated at $\hat{s}[n | n - 1]$ is

$$H[n] = \left. \frac{\partial h(s)}{\partial x} \right|_{s = \hat{s}[n | n - 1]}$$

Let

$$\rho = \sqrt{r_x^2 + r_y^2}$$

Then the Jacobian becomes

$$H[n] = \begin{bmatrix} \frac{r_x}{\rho} & \frac{r_y}{\rho} & 0 & 0 \\ -\frac{r_y}{\rho^2} & \frac{r_x}{\rho^2} & 0 & 0 \end{bmatrix}$$

Update:

Innovation: $e[n] = x[n] - h(\hat{s}[n | n - 1])$

Innovation covariance: $S[n] = H[n]M[n | n - 1]H^T[n] + C$

Kalman gain: $K[n] = M[n | n - 1]H^T[n]S^{-1}[n]$

Corrected estimate: $\hat{s}[n | n] = \hat{s}[n | n - 1] + K[n]e[n]$

Corrected Covariance: $M[n | n] = (I - K[n]H[n])M[n | n - 1]$

The EKF covariance matrix provides an estimate of the uncertainty in the state estimate. In particular, the diagonal elements $M_{11}[n|n]$ and $M_{22}[n|n]$ represent the EKF's estimated minimum MSE for $r_x[n]$ and $r_y[n]$, respectively, and can be plotted to illustrate filter convergence and time-varying uncertainty.

Results and Discussion

Task 1:

Let $\sigma_u^2 = 0.0001$, $\sigma_r^2 = 0.1$, $\sigma_b^2 = 0.01$. Implement the extended Kalman filter to estimate the position of the vehicle with initial state $\hat{s}[-1|-1] = [5500]^T$. Plot the figures for the estimated position vs true one and the minimum MSE versus samples.

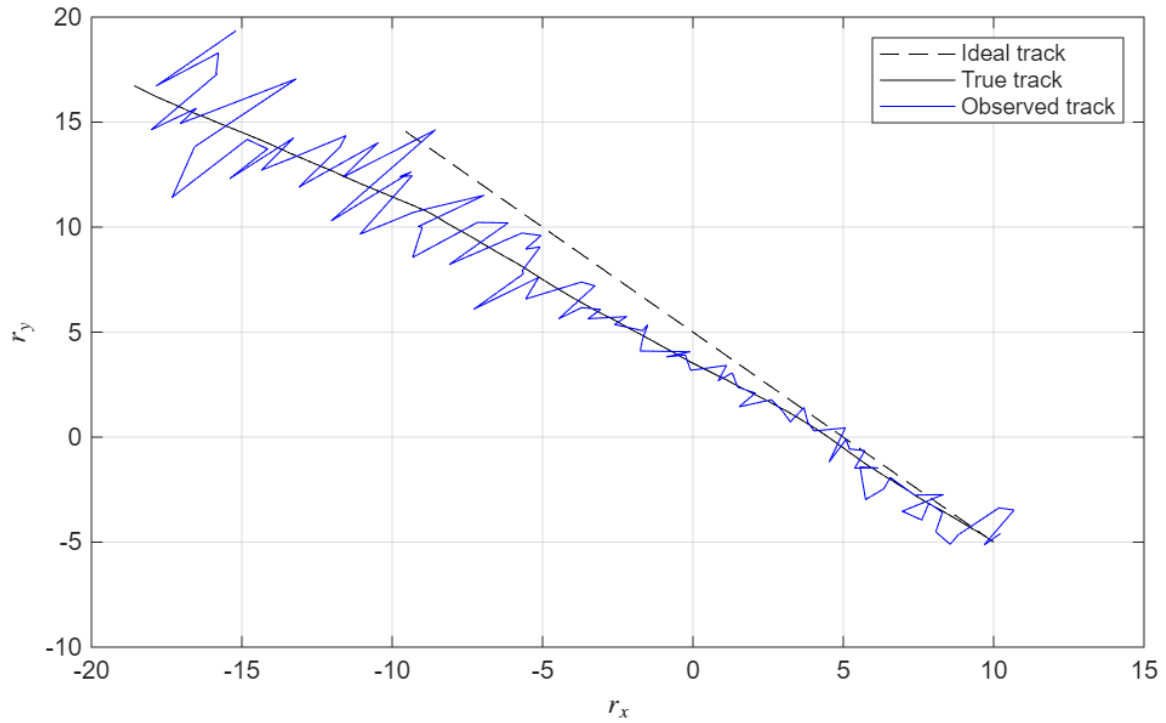


Fig 1.1. Ideal track, True track, and Observed track

Fig 1.1. shows the vehicle trajectory in the Cartesian plane. The ideal track (dashed line) represents the nominal straight-line motion assuming constant velocity. The true track deviates slightly from the ideal path due to the added process noise (driving noise), which models random velocity perturbations. The observed track is obtained from noisy range and bearing measurements converted back into Cartesian coordinates and therefore shows significant fluctuations around the true track due to measurement noise.

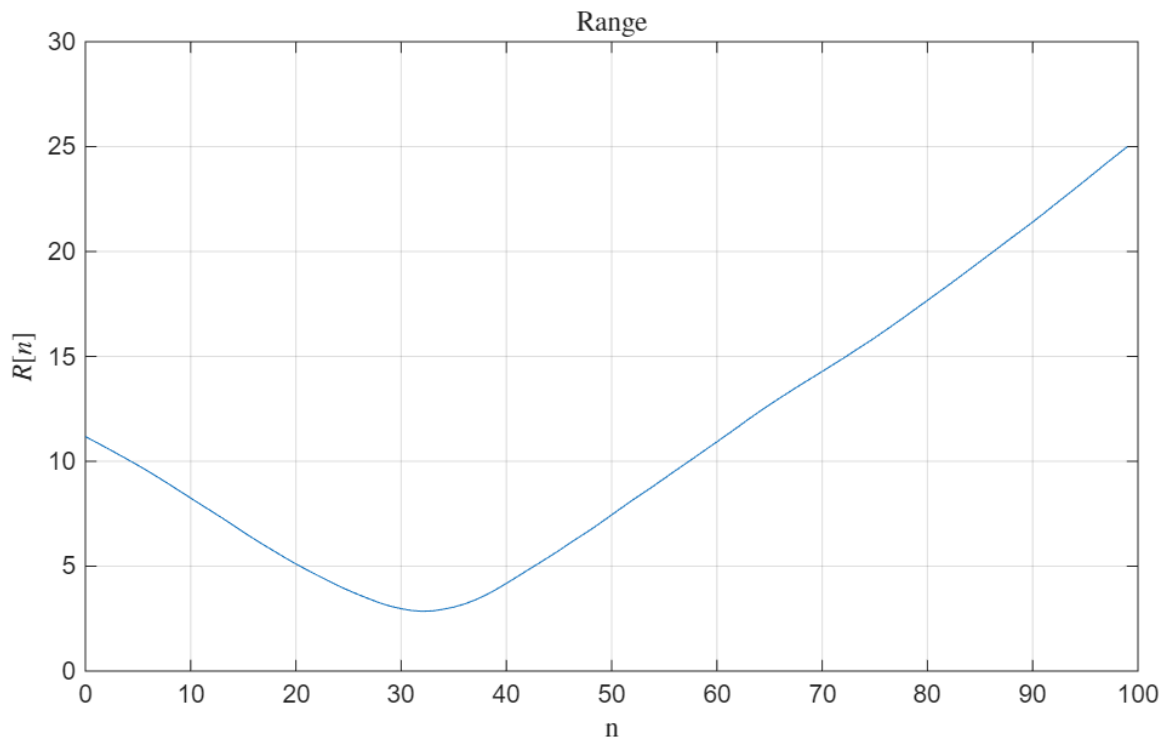


Fig 1.2. Range $R[n]$

This plot fig 1.2 shows the true range between the vehicle and the origin as a function of time. Initially, the range decreases as the vehicle approaches the origin, reaching a minimum around $n \approx 30$. After this point, the range increases as the vehicle moves away from its origin. The smooth shape indicates that the vehicle follows an approximately straight trajectory, consistent with the constant velocity model.

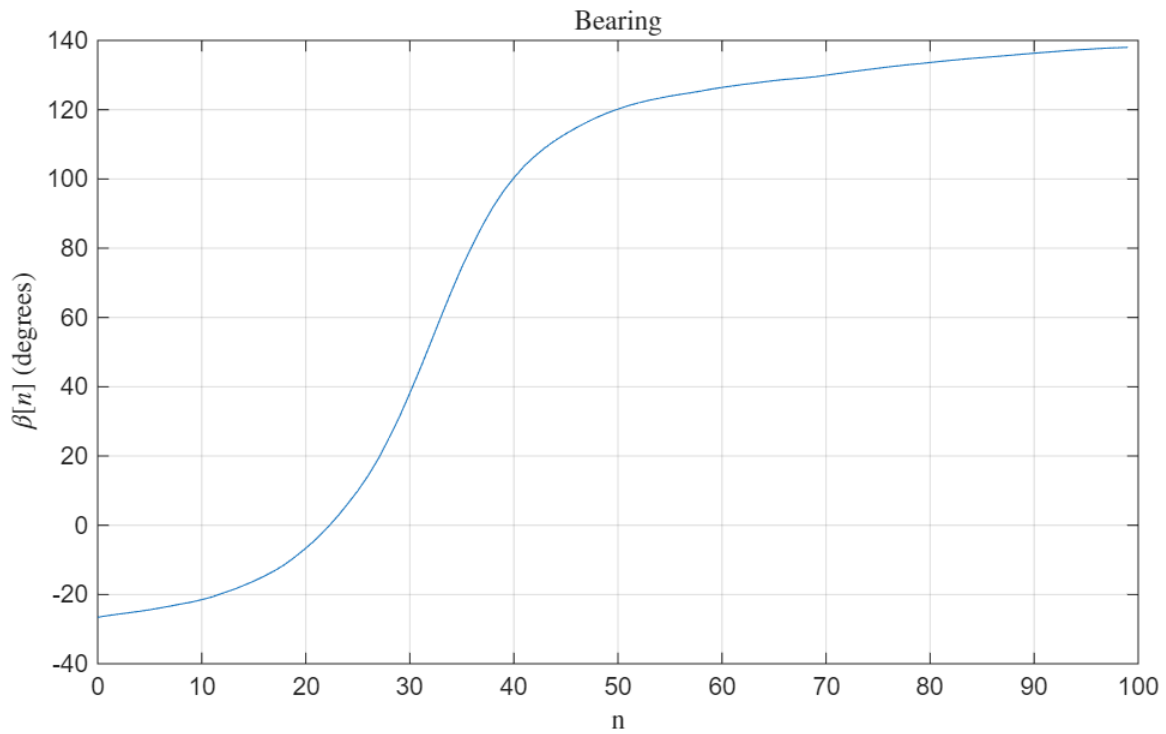


Fig 1.3. Bearing $\beta[n]$

This figure 1.3 illustrates the evolution of the bearing angle over time. The bearing increases steadily from negative angles to positive angles as the vehicle moves across the observation region. The rapid change in bearing around $n \approx 30$ corresponds to the vehicle passing closest to the origin, where small position changes cause larger angular variations. This nonlinear behavior motivates the use of an EKF instead of a standard Kalman filter.

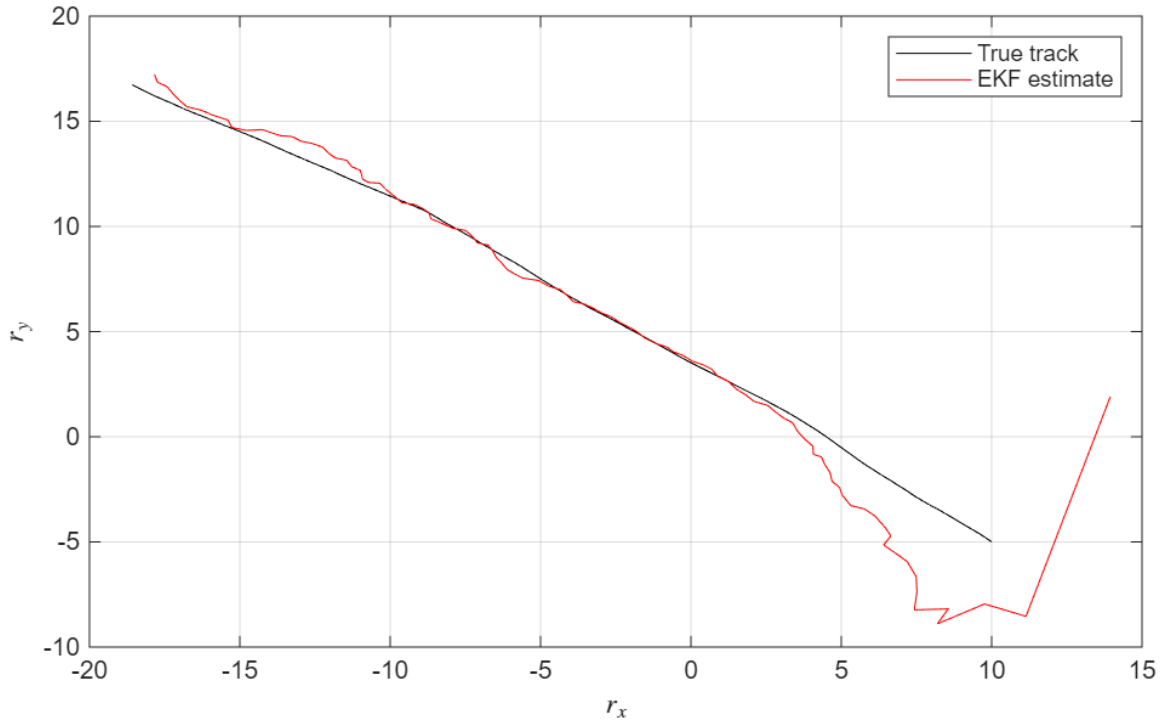


Fig 1.4. True track vs EKF estimate

In figure 1.4 EKF estimated trajectory is compared against the true vehicle track. The EKF estimate initially differs from the true trajectory due to the poor initial state guess, but it converges after several samples as more measurements are incorporated. The EKF follows the true track closely for most of the trajectory. However, near the end of the track, the estimate shows some divergence and oscillation, which is expected since EKF performance depends on linearization accuracy and can degrade when the model becomes sensitive to nonlinear measurement effects.

Figures 1.5 and 1.6 show the EKF-estimated minimum mean-square error (minimum MSE) for the position components $r_x[n]$ and $r_y[n]$, given by the covariance terms $M_{11}[n]$ and $M_{22}[n]$, respectively. At the beginning, both curves exhibit large values because the filter is initialized far from the true state and with a large initial uncertainty ($M[-1 | -1] = 100I$). As more measurements are processed, the EKF converges and the uncertainty decreases rapidly, reaching a minimum after the transient phase. After convergence, both MSE curves gradually increase again because the process (driving) noise continuously injects uncertainty into the velocity states, which accumulates into the position estimates over time. The initial peak is higher for $r_y[n]$ than for $r_x[n]$, indicating that the measurement geometry and nonlinear range-bearing mapping led to greater uncertainty in the y-direction during the initial convergence phase.

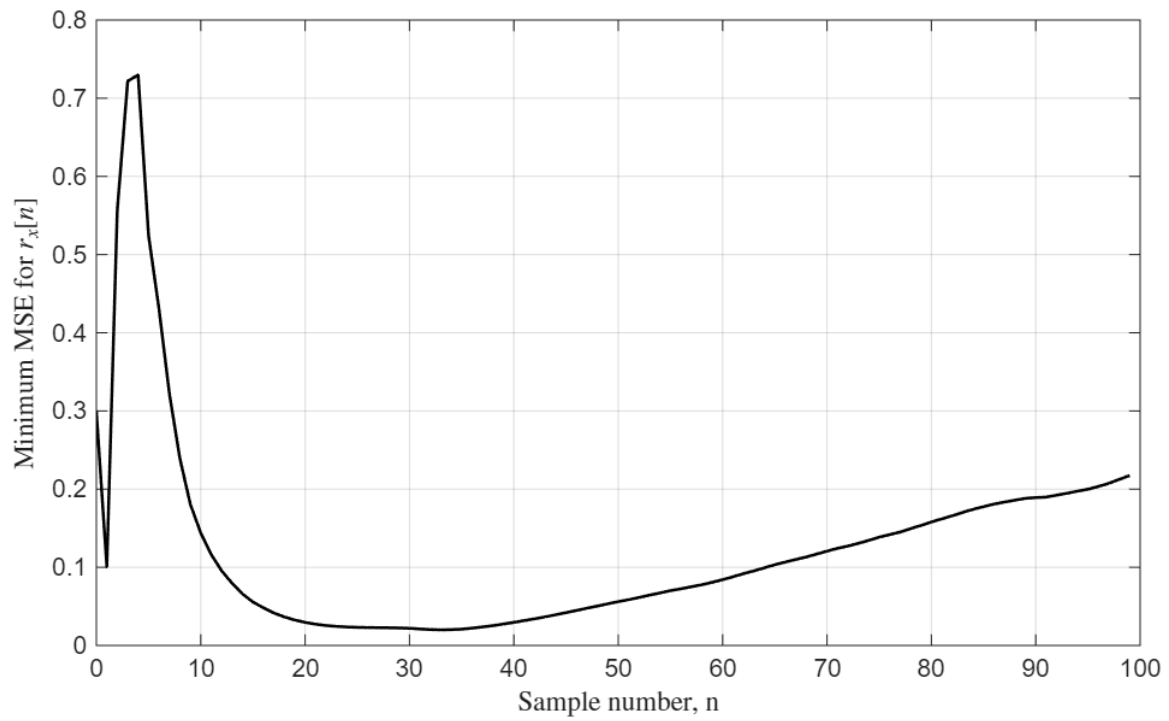


Fig 1.5. Minimum MSE vs sample

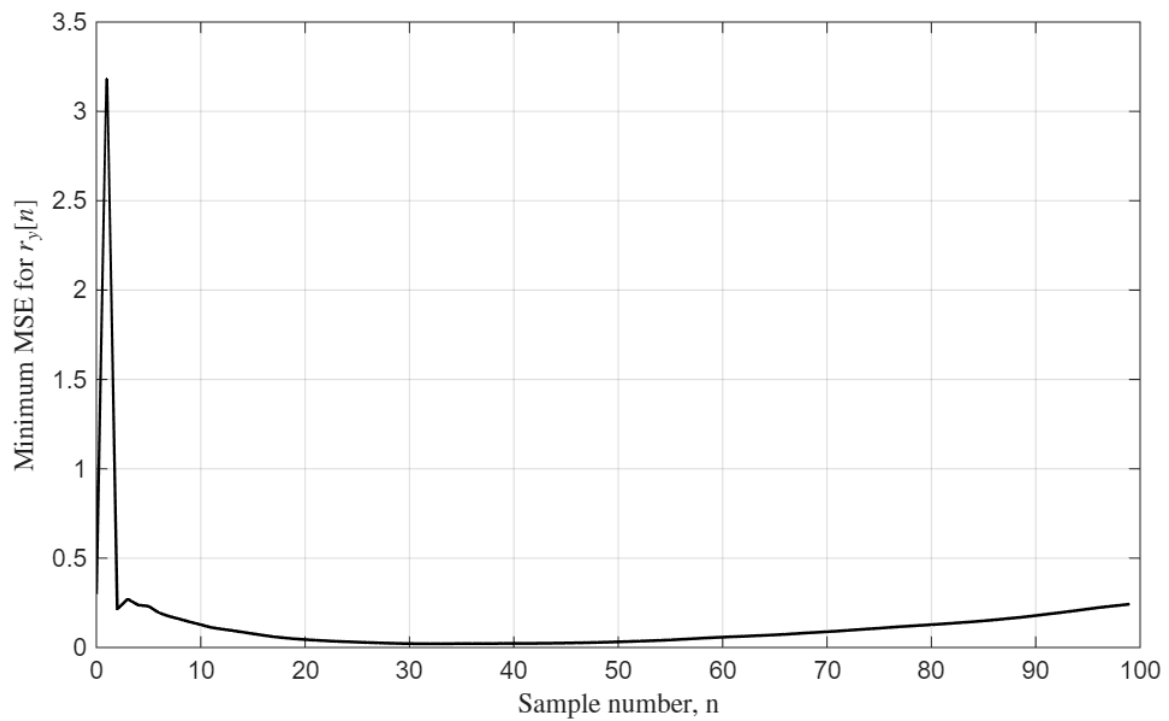


Fig 1.6. Minimum MSE vs sample

Task 2: Repeat task 1) with a different initial state. Give your comments on the performance of the extended Kalman filter.

While changing the initial EKF state estimate to $\hat{s}[-1 | -1] = [50 \ -50 \ 2 \ -2]^T$, the filter starts extremely far from the true vehicle position and velocity. As a result, the EKF linearization is performed around a highly incorrect predicted state, causing the filter update step to become unreliable. This leads to strong divergence in the estimated trajectory, as seen in the tracking plot where the EKF estimate deviates significantly from the true track and produces unrealistic motion. The minimum MSE plots for $r_x[n]$ and $r_y[n]$ show very large uncertainty at the beginning, reflecting the poor initialization, but then rapidly decrease because the EKF covariance update reduces uncertainty after incorporating measurements. However, despite the covariance shrinking, the state estimate remains incorrect, indicating inconsistency caused by EKF sensitivity to large initialization errors and nonlinear measurement effects.

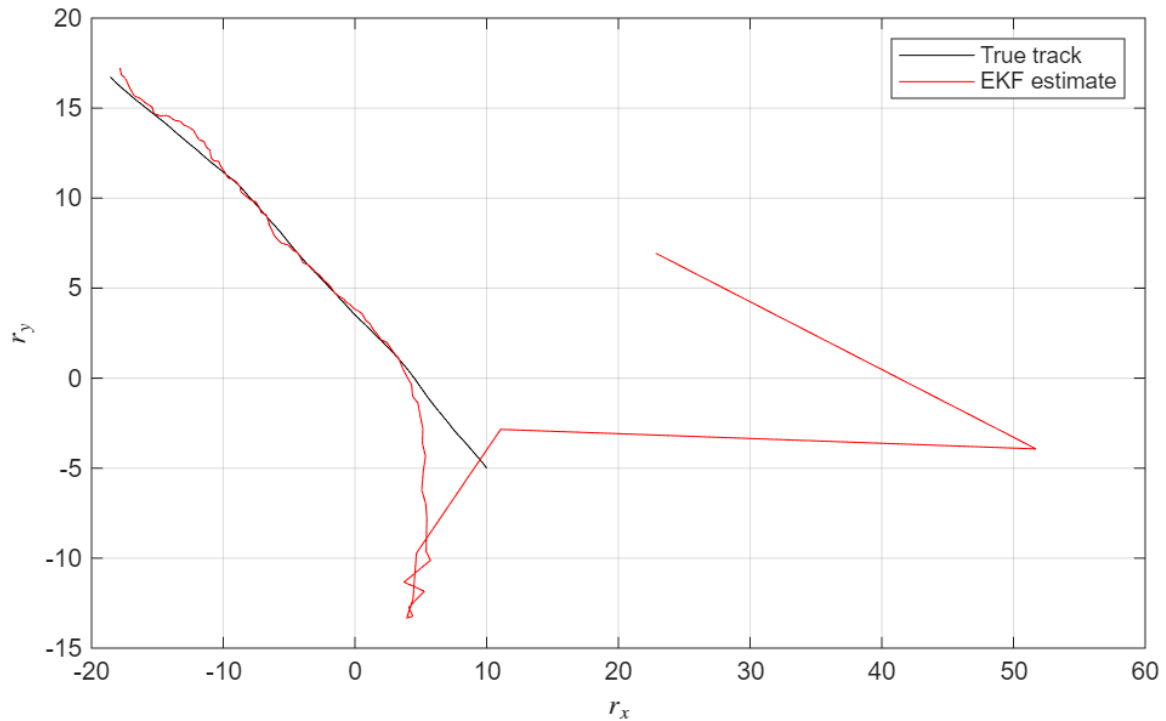


Fig 2.1. True track vs EKF estimate

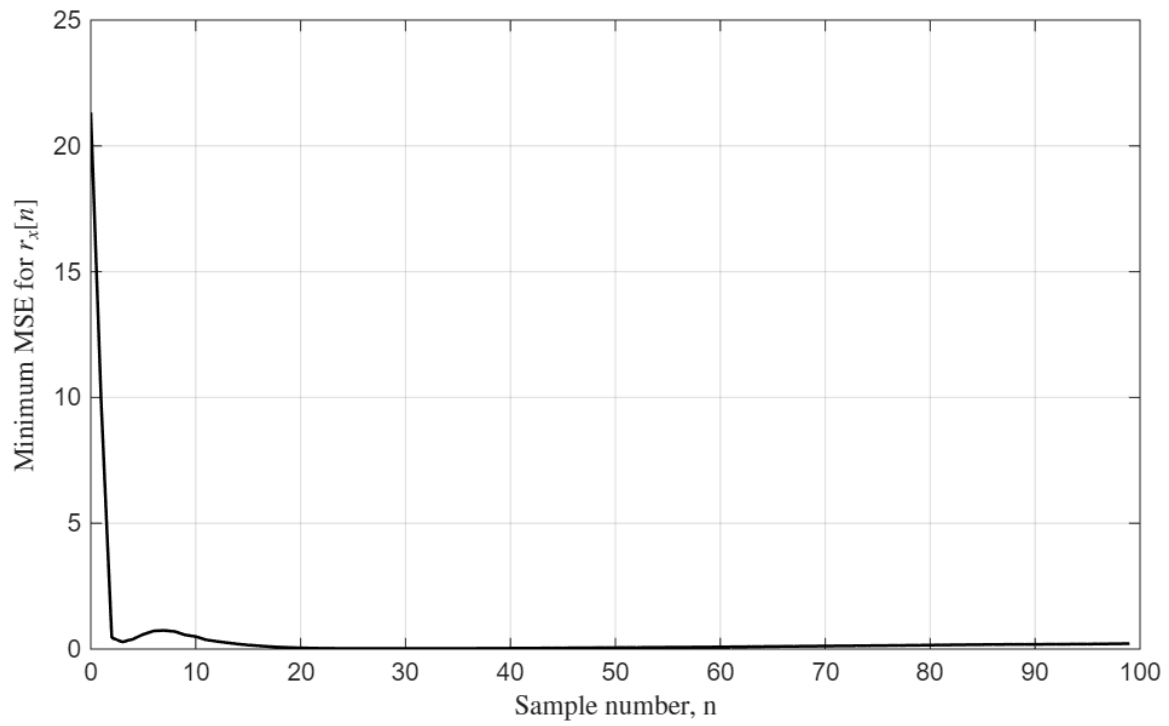


Fig 2.2. Minimum MSE vs sample

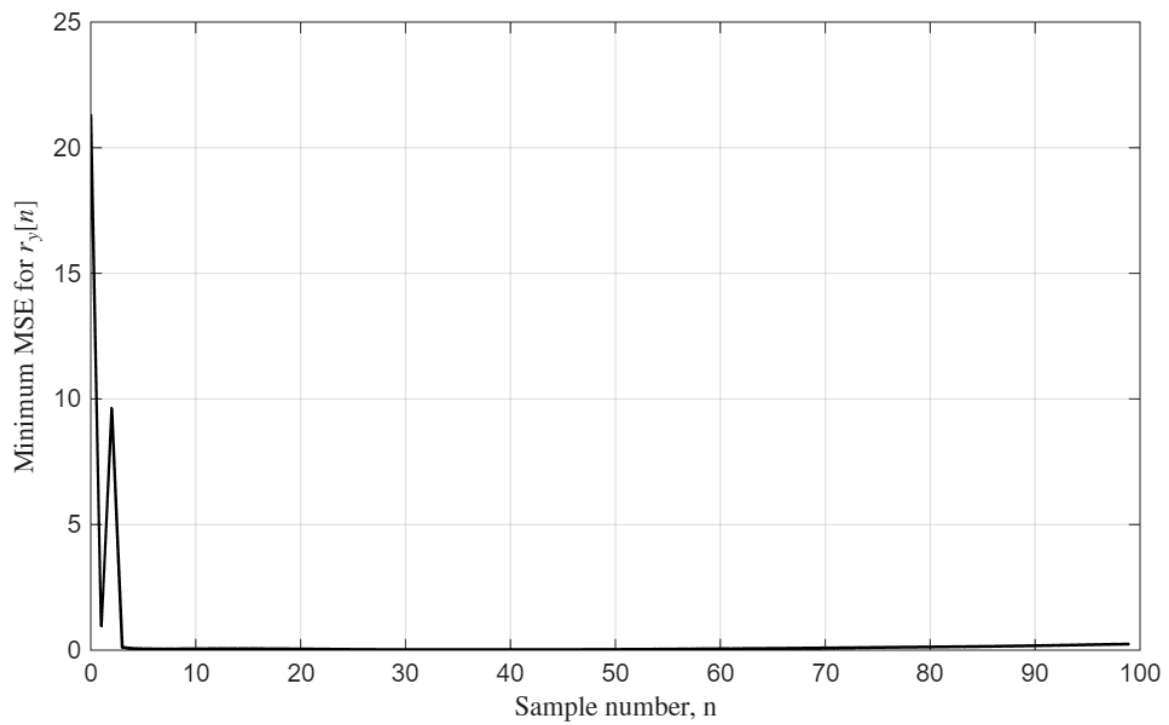


Fig 2.3. Minimum MSE vs sample

Although the minimum MSE decreases rapidly, the state estimate remains far from the true trajectory. This indicates filter inconsistency, caused by the EKF linearizing the nonlinear measurement model around a severely incorrect initial estimate. As a result, covariance underestimates the true estimation error.

Task 3: Let $\sigma_u^2 = 0.01$, $\sigma_r^2 = 0.1$, $\sigma_b^2 = 0.01$. Repeat task 1) and give your comments.

When the driving noise variance is increased to $\sigma_u^2 = 0.01$, the vehicle velocity becomes much more unstable, meaning the target no longer follows a smooth straight-line motion. This effect is clearly visible in fig 3.1, where the true trajectory deviates more from the ideal dashed line and the observed track shows larger fluctuations. Since the vehicle passes close to the origin, the bearing plot exhibits a sharp jump around $n \approx 15$, which is expected because the bearing angle becomes highly sensitive when the range is small. In the EKF trajectory plot, the estimate follows the true track reasonably well at the beginning but later shows noticeable deviation and instability near the end, indicating that the filter struggles to maintain accurate tracking under strong process noise. This behavior is also reflected in the minimum MSE plots, where both $M_{11}[n]$ and $M_{22}[n]$ increase significantly over time instead of remaining small, demonstrating that the EKF predicts growing uncertainty in the position estimate as the process noise continuously accumulates into the state dynamics. Overall, the graphs confirm that a larger σ_u^2 leads to higher trajectory randomness, noisier observations, and increasing estimation uncertainty.

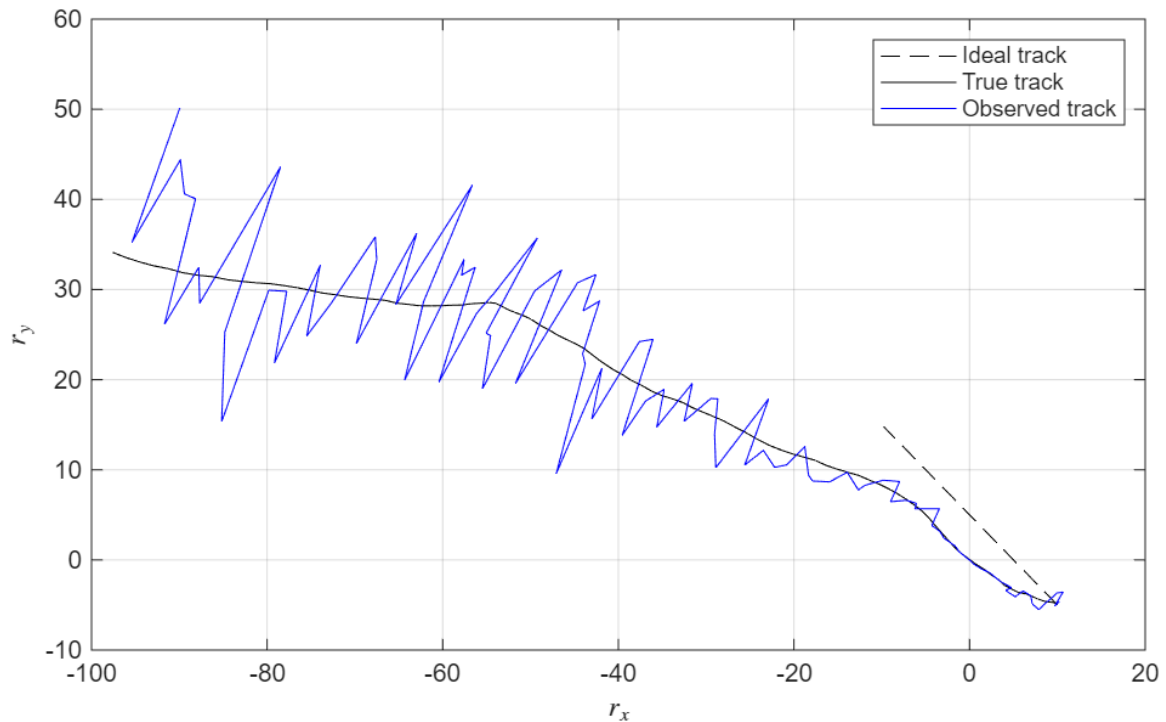


Figure 3.1 Ideal track, True track, and Observed track

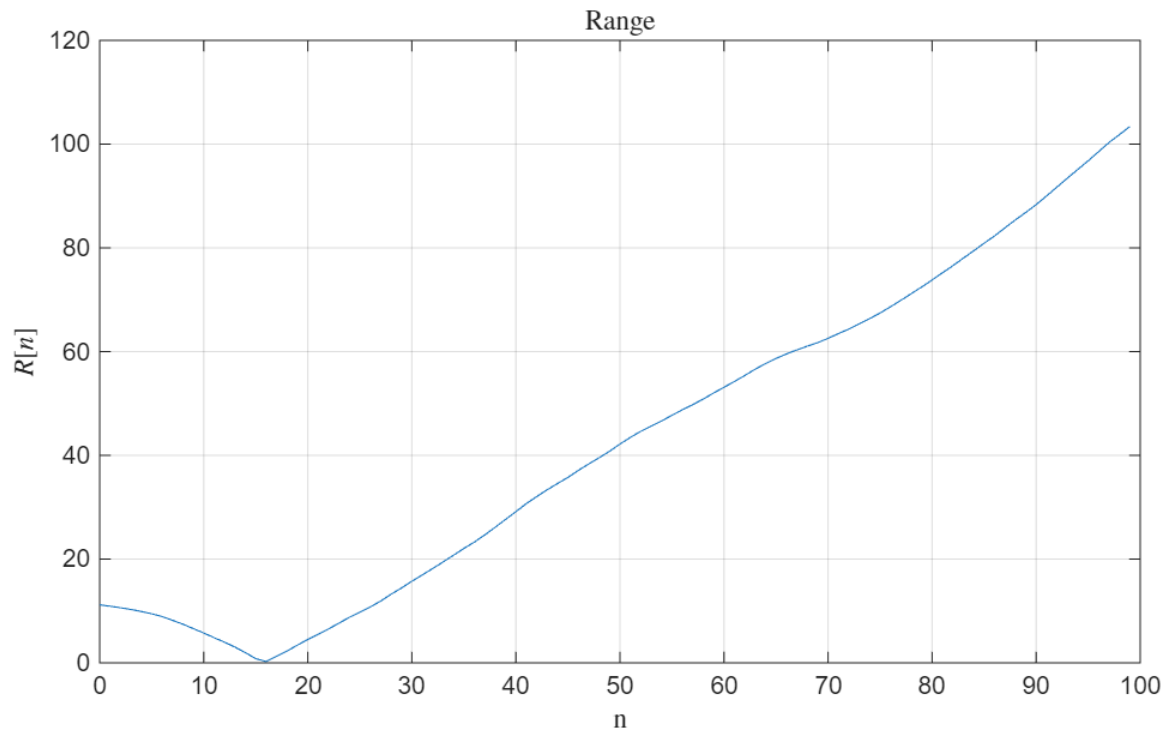


Figure 3.2 Range $R[n]$

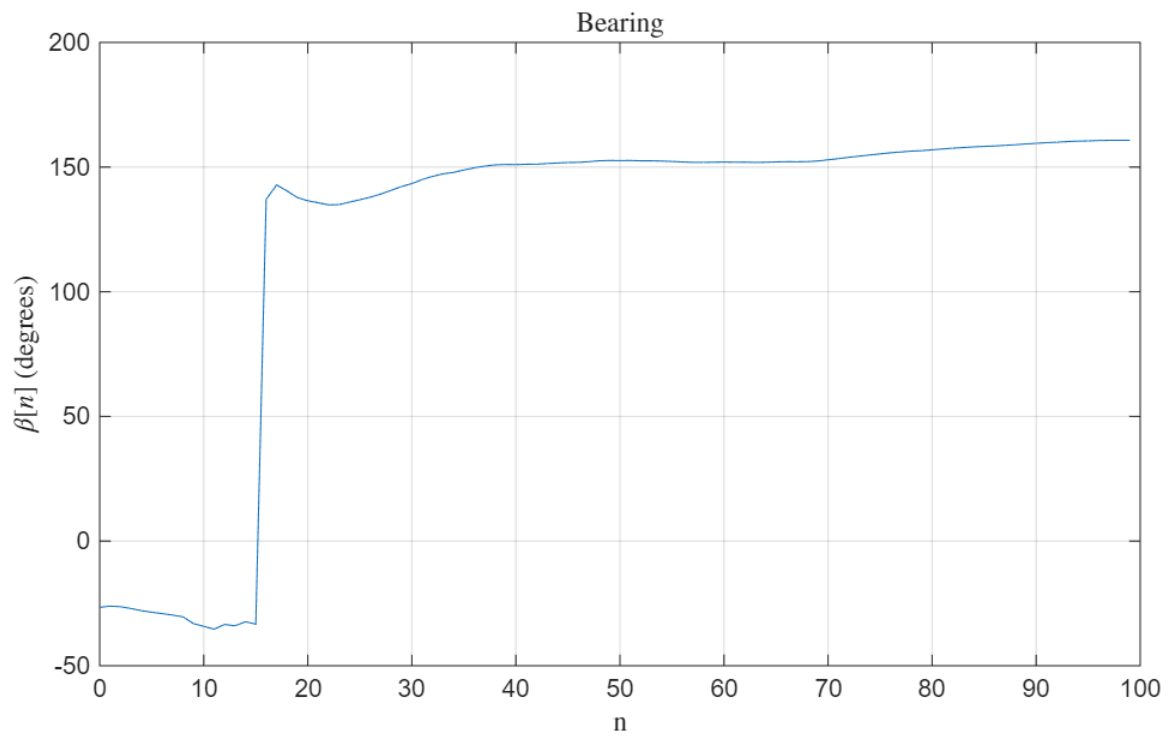


Figure 3.3 Bearing $\beta[n]$

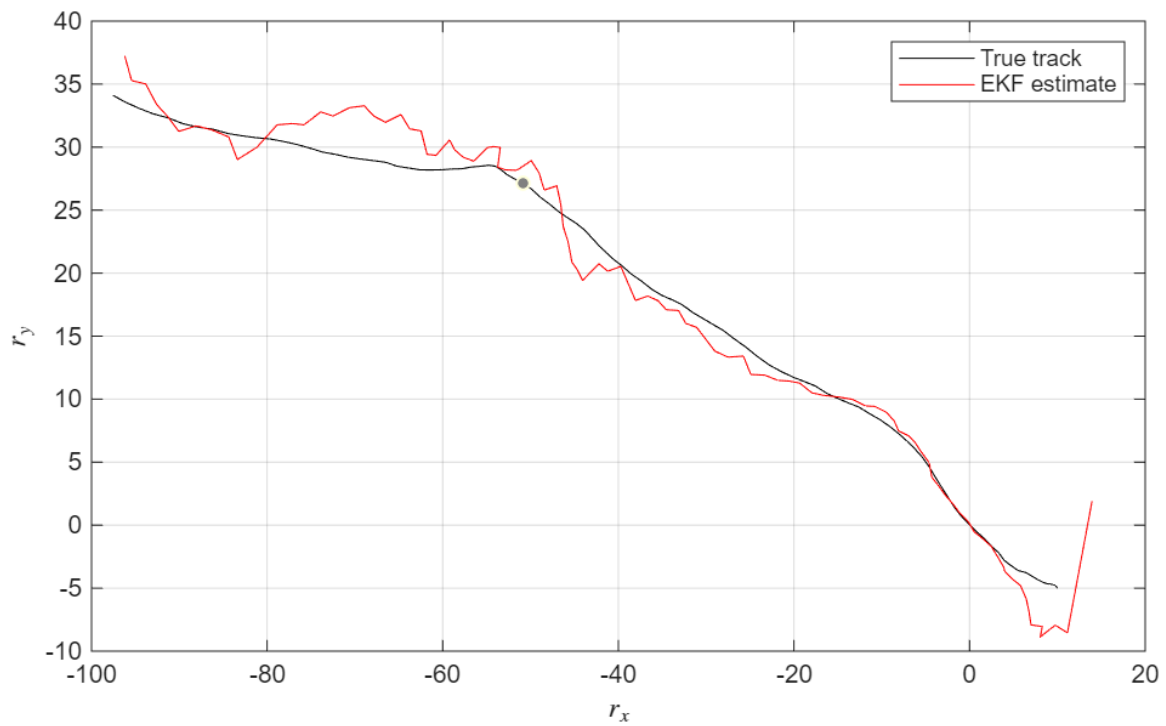


Figure 3.4 True track vs EKF estimate

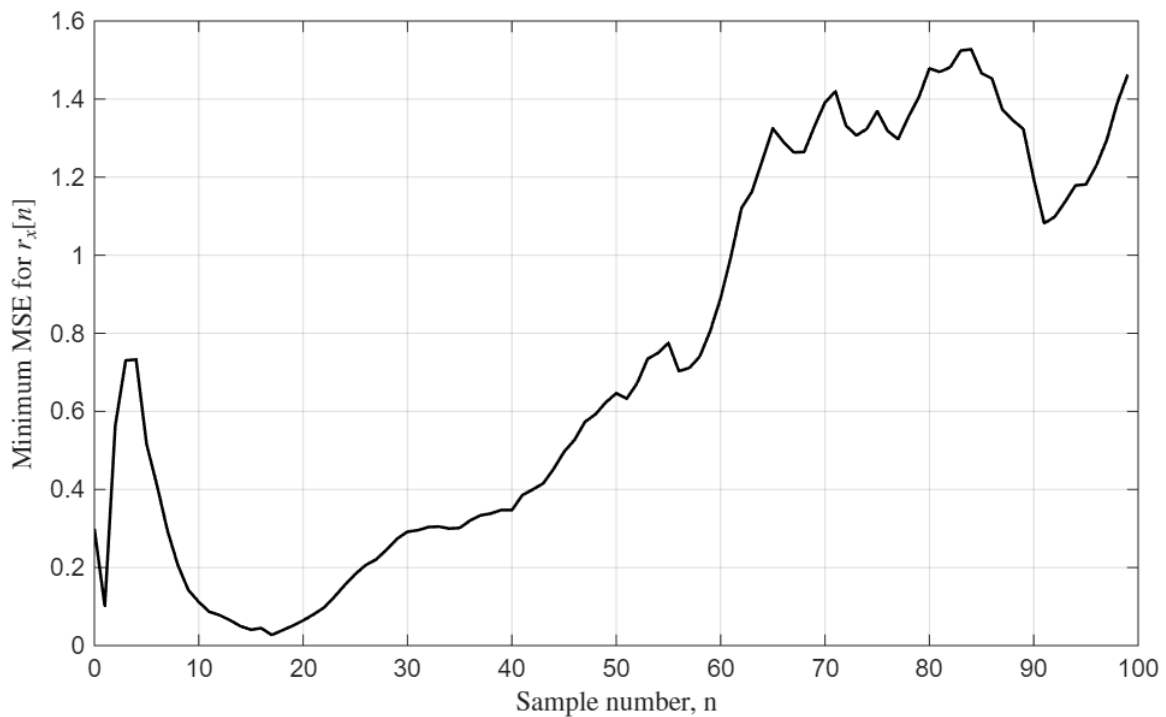


Figure 3.5 Minimum MSE vs sample

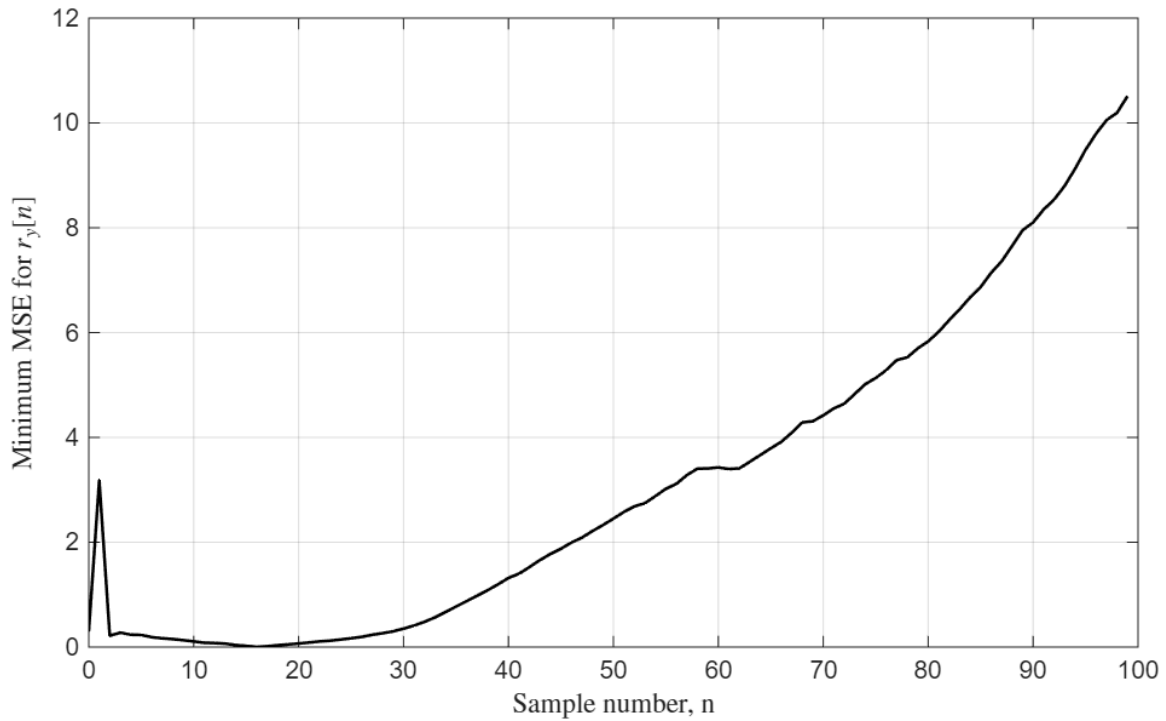


Figure 3.6 Minimum MSE vs sample

Task 4: Let $\sigma_u^2 = 0.0001$, $\sigma_r^2 = 1$, $\sigma_b^2 = 0.5$. Repeat task 1) and give your comments.

Here, the process noise variance is kept very small ($\sigma_u^2 = 0.0001$), meaning the motion model assumes nearly constant velocity, while the measurement noise variances are increased significantly ($\sigma_r^2 = 1$, $\sigma_b^2 = 0.5$). As seen in fig 4.1, the observed track becomes highly scattered due to the strong measurement noise, especially in bearing (fig 4.3), which produces large fluctuations in the Cartesian conversion. However, since the process noise is small, the EKF relies more heavily on the motion model and less on the noisy measurements. This is reflected in the tracking plot, where the EKF estimate remains relatively smooth and follows the general direction of the true trajectory, although some deviation appears toward the end due to accumulated measurement uncertainty. The minimum MSE plots initially decrease as the filter converges but later increase gradually, indicating that despite trusting the motion model, the high measurement noise limits the achievable estimation accuracy. Overall, increasing measurement noise makes the observations unreliable, causing larger estimation uncertainty and reduced tracking precision compared to Task 1.

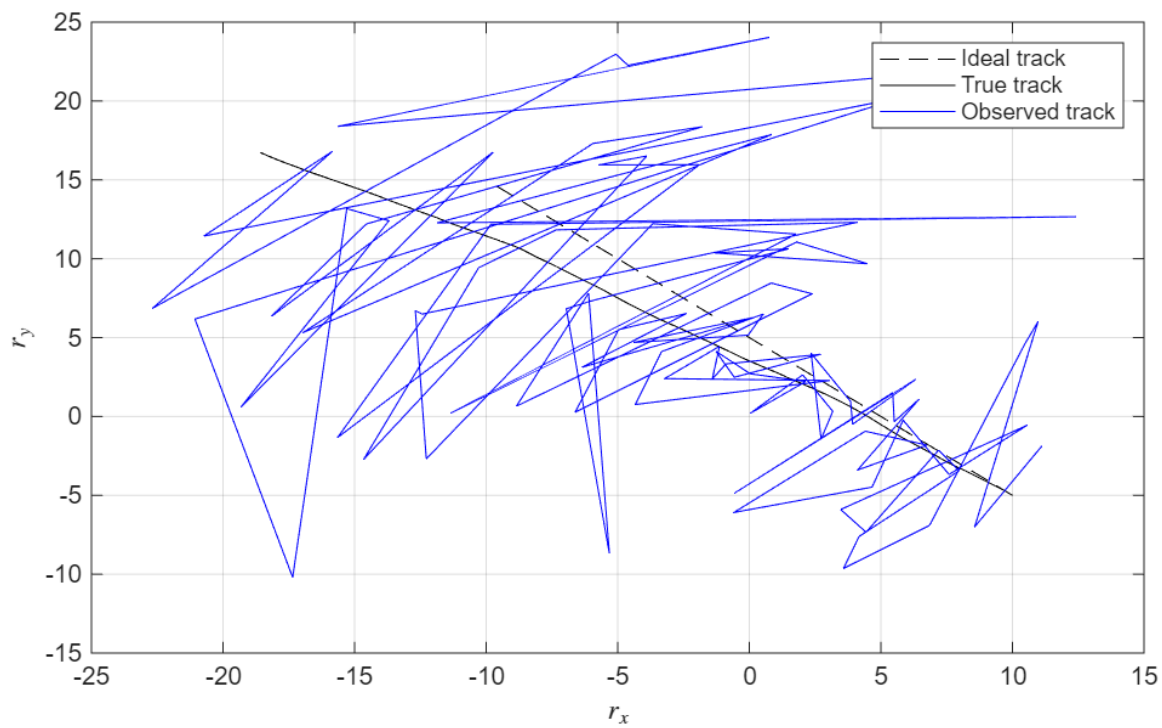


Figure 4.1 Ideal track, True track, and Observed track

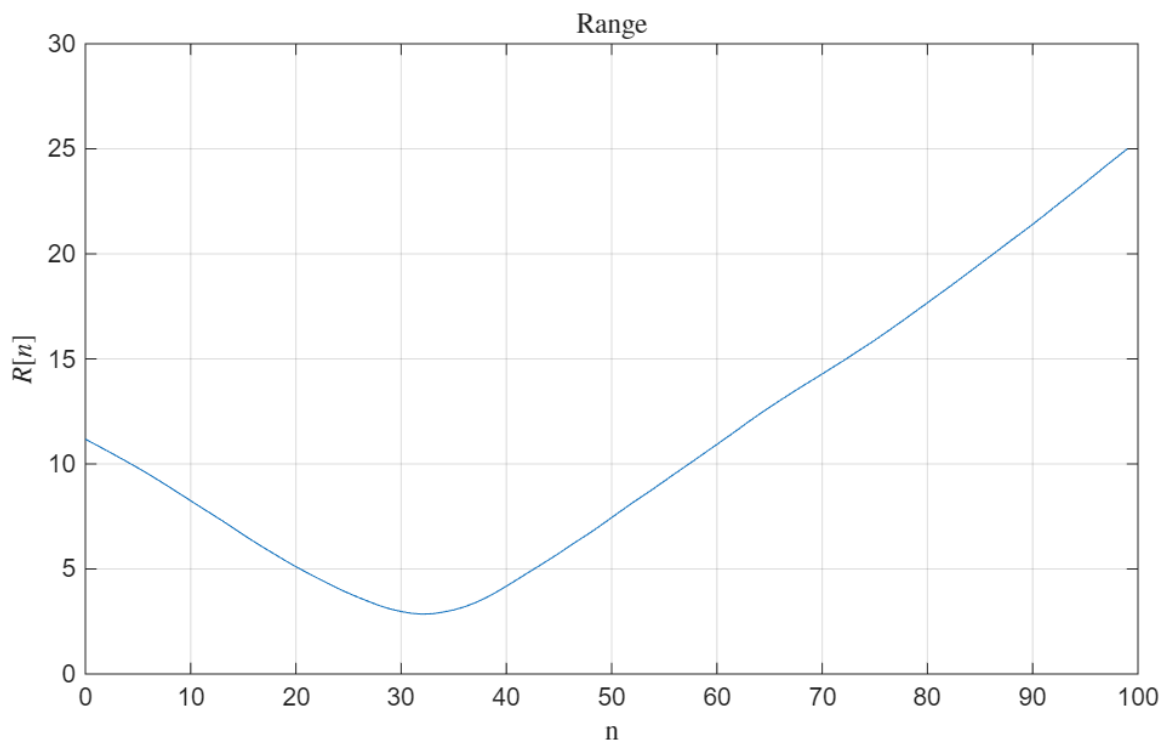


Figure 4.2 Range $R[n]$

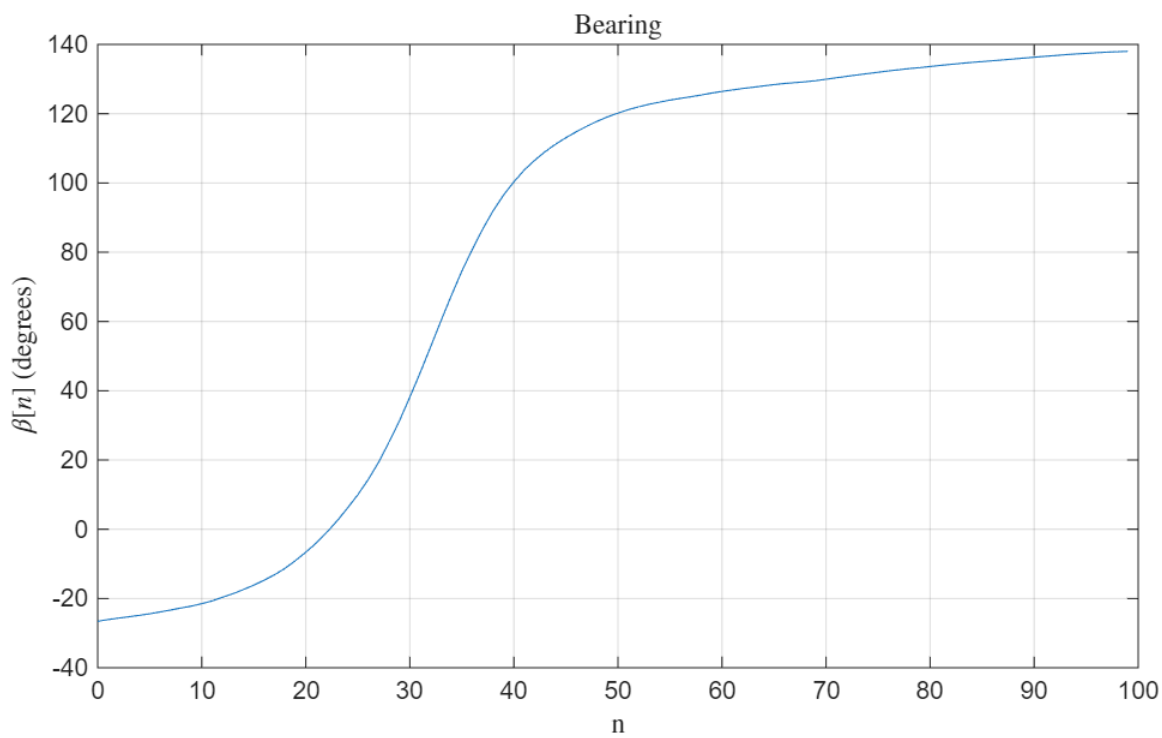


Figure 4.3 Bearing $\beta[n]$

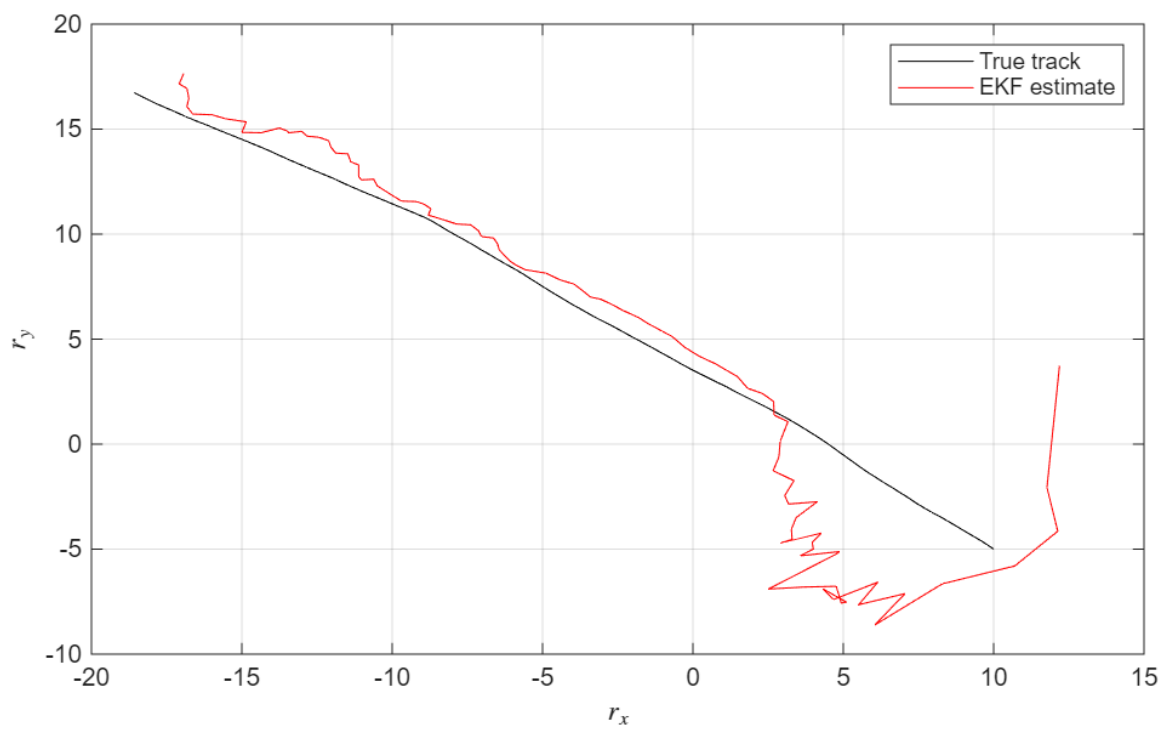


Figure 4.4 True track vs EKF estimate

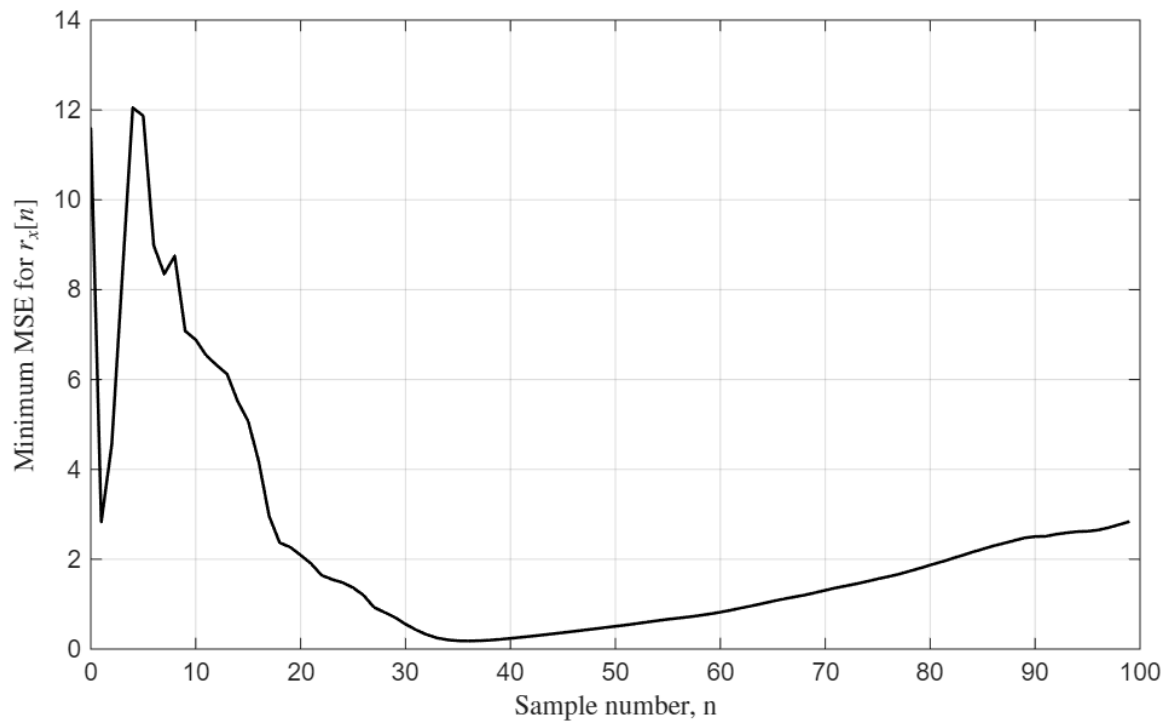


Figure 4.5 Minimum MSE vs sample

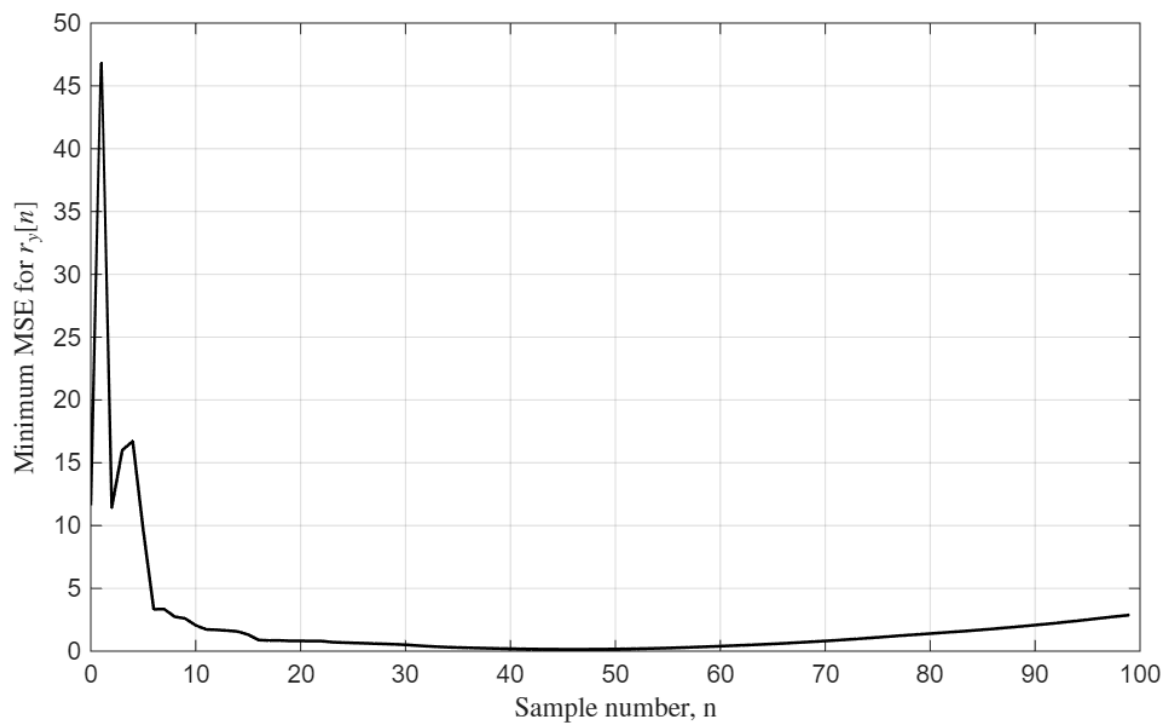


Figure 4.6 Minimum MSE vs sample

Conclusion

In this Lab, the Extended Kalman Filter (EKF) was successfully implemented to estimate the position and velocity of a moving vehicle using nonlinear range and bearing measurements. The results of Task 1 matched the reference example from the textbook, confirming the correctness of the EKF prediction and update equations. EKF was able to converge from an inaccurate initial state estimate and provided a trajectory close to the true vehicle track, while the minimum MSE curves demonstrated the expected decrease during convergence and gradual increase due to accumulated process noise.

In the later experiments we have observed that EKF performance is highly sensitive to initialization and noise statistics. When the initial estimate was chosen far from the true state, EKF produced large tracking errors even though the covariance decreased, illustrating filter inconsistency caused by incorrect linearization. Increasing the process noise variance resulted in larger deviations of the true trajectory and increased uncertainty in the EKF estimates, as reflected by higher minimum MSE values. On the other hand, increasing measurement noise significantly degraded the observation quality, forcing the EKF to rely more on the motion model and producing smoother but less accurate estimates. Overall, the assignment demonstrated both the effectiveness and limitations of the EKF, highlighting that accurate initialization and appropriate noise modeling are crucial for stable and reliable nonlinear tracking performance.

References

- [1] S. M. Kay, Fundamentals of Statistical Signal Processing: Estimation Theory. Prentice Hall International, 1993.
- [2] K. P. Murphy, Machine Learning: A Probabilistic Perspective. Cambridge, MA, USA: MIT Press, 2012, Sec. 15.2.